

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
Higher 1

CANDIDATE  
NAME

**ANSWERS**

CLASS

TUTOR'S  
NAME

**CHEMISTRY**

Paper 2

**8872/02**

**19 September 2013**

**2 hours**

Candidates answer Section A on the Question Paper.

Additional Materials:      Answer Paper  
                                    Data Booklet  
                                    Graph Paper

**READ THESE INSTRUCTIONS FIRST**

Write your name and class on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A**

Answer **all** the questions.

**Section B**

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| Section A          |  |
| B5                 |  |
| B6                 |  |
| B7                 |  |
| Total              |  |

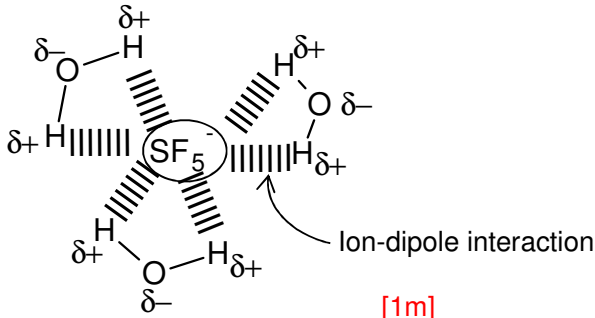
This document consists of **16** printed pages.

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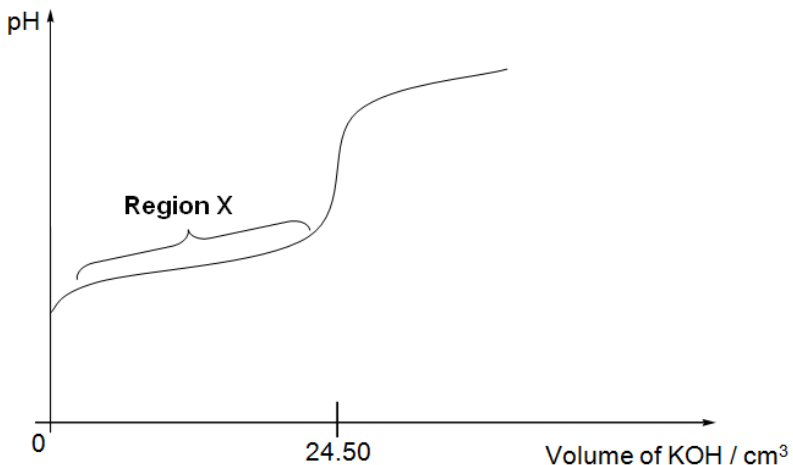
## Section A

Answer **all** the questions in this section in the spaces provided.

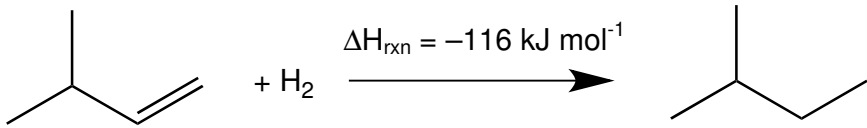
|   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|---|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | (a)  | Silicon and sulfur are among the most abundant period 3 elements on Earth.                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|   | (i)  | Describe the variations in melting points of the elements silicon and sulfur. Explain the variations in terms of their structures and bonding.                                                                                                                                                                                                                                                                                                                                                                           |  |
|   |      | <p>Silicon has <u>giant molecular structure held together by numerous strong covalent bonds between atoms</u>. As large amount of energy is required to break the strong covalent bonds, silicon has a higher melting point. [1m]</p> <p>Sulfur has <u>simple molecular structure with weak van der Waals forces of attractions between molecules</u>. Lesser amounts of energy is required to break the weak van der Waals forces of attraction hence sulfur has lower melting point. [1m]</p>                          |  |
|   | (ii) | Describe the action of water on the oxides of silicon and sulfur, write equations for any reactions that occur, and suggest the pH of each solution formed. [5]                                                                                                                                                                                                                                                                                                                                                          |  |
|   |      | <p>SiO<sub>2</sub> is insoluble in water because energy from the weak van der Waals forces between SiO<sub>2</sub> and H<sub>2</sub>O molecules are unable to overcome the strong covalent bonds between Si and O atoms (pH = 7). [1m]</p> <p>Oxides of sulfur reacts with water to give a highly acidic solution (pH = 1-2). [1m]</p> <p>SO<sub>2</sub> (g) + H<sub>2</sub>O (l) → H<sub>2</sub>SO<sub>3</sub> (aq) [1m for both]</p> <p>SO<sub>3</sub> (g) + H<sub>2</sub>O (l) → H<sub>2</sub>SO<sub>4</sub> (aq)</p> |  |
|   | (b)  | Sulfur hexafluoride (SF <sub>6</sub> ) is a colourless and odourless gas that is poorly soluble in water. It is hypothesised that sulfur pentafluoride ion (SF <sub>5</sub> <sup>-</sup> ) is very soluble in water.                                                                                                                                                                                                                                                                                                     |  |
|   | (i)  | Draw a dot-and-cross diagram to show the bonding in a sulfur pentafluoride ion.                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |

|  |       |                                                                                                                                                                                                                                                                                                                                  |
|--|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | (ii)  | State and explain in detail how the shape of sulfur pentafluoride ion arises.                                                                                                                                                                                                                                                    |
|  |       | There are 5 bond pairs and 1 lone pair of electrons around the central sulfur atom. [1m]<br>Hence shape is square pyramidal. [1m]                                                                                                                                                                                                |
|  | (iii) | Explain with the aid of a diagram why sulfur pentafluoride ion is very soluble in water. [5]                                                                                                                                                                                                                                     |
|  |       | <p><math>\text{SF}_5^-</math> is a molecular ion. The ion-dipole interaction between the ion and water is strong enough to overcome the strong hydrogen bonds between water molecules. [1m] Hence it is very soluble in water.</p>  <p>[1m]</p> |
|  |       | [Total: 10]                                                                                                                                                                                                                                                                                                                      |

|                       |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |              |      |                       |      |            |      |
|-----------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|--------------|------|-----------------------|------|------------|------|
| 2                     | (a)  | <p>Given the following information, explain why the different compounds have different pKa values at 298 K.</p> <table><tr><td>Compound</td><td>pKa</td></tr><tr><td>Benzoic acid</td><td>4.19</td></tr><tr><td>2-methylbutanoic acid</td><td>4.80</td></tr><tr><td>Butan-2-ol</td><td>17.6</td></tr></table> <p>[4]</p>                                                                                                                                                                                                                                                                                                                                                                  | Compound | pKa | Benzoic acid | 4.19 | 2-methylbutanoic acid | 4.80 | Butan-2-ol | 17.6 |
| Compound              | pKa  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |              |      |                       |      |            |      |
| Benzoic acid          | 4.19 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |              |      |                       |      |            |      |
| 2-methylbutanoic acid | 4.80 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |              |      |                       |      |            |      |
| Butan-2-ol            | 17.6 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |              |      |                       |      |            |      |
|                       |      | <p>The strength of an acid depends on the stability of the conjugate base. The more stable the conjugate base, the stronger the acid.</p> <p>The higher the pKa, the lower the Ka, the weaker the acid. Hence benzoic acid is the strongest acid.</p> <p>Hence, the order of stability of the conjugate base is benzoate ion &gt; 2-methylbutanoate ion &gt; butoxide ion [1m]</p> <p>In <math>\text{CH}_3(\text{CH}_2)_3\text{O}^-</math>, the presence of electron-donating <math>-(\text{CH}_2)_3\text{CH}_3</math> group intensifies the negative charge on oxygen. Hence <math>\text{CH}_3(\text{CH}_2)_3\text{O}^-</math> is destabilised. Butan-2-ol is the weakest acid. [1m]</p> |          |     |              |      |                       |      |            |      |

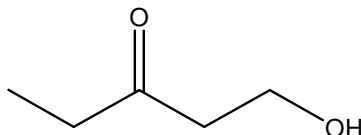
|  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |       | <p>In <math>\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2^-</math> and <math>\text{C}_6\text{H}_5\text{CO}_2^-</math>, the p-orbital of <math>\text{O}^-</math> overlaps with the p orbitals of the <math>\text{C}=\text{O}</math> bond. Hence the negative charge is effectively dispersed over one carbon atom and two electronegative oxygen atoms. [1m] However, <math>\text{C}_6\text{H}_5\text{CO}_2^-</math> is more stabilised than <math>\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2^-</math> as the negative charge can be dispersed into benzene (more effectively) unlike <math>\text{CH}_3\text{CO}_2^-</math>. Benzoic acid is therefore a stronger acid than ethanoic acid and stronger than ethanol. [1m]</p> |
|  | (b)   | <p>25.0 <math>\text{cm}^3</math> of 2-methylbutanoic acid was titrated against 0.15 <math>\text{mol dm}^{-3}</math> of KOH and the following titration curve was obtained.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | (i)   | <p>Explain what is meant by the term acid dissociation constant, <math>K_a</math>, of 2-methylbutanoic acid.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  |       | <p>The acid dissociation constant is defined as <math>K_a</math></p> $= \frac{[\text{H}^+][\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2^-]}{[\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2\text{H}]} \quad [1\text{m}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | (ii)  | <p>Calculate the concentration of 2-methylbutanoic acid used in this experiment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |       | $[\text{acid}] = \frac{(0.15)(24.50)}{(25.0)} = 0.1470 \text{ mol dm}^{-3} \quad [1\text{m}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | (iii) | <p>Using relevant values from (a) and b (ii), calculate the initial pH of 2-methylbutanoic acid acid.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  |       | <p>Given <math>\text{p}K_a = 4.80</math>:</p> $[\text{H}^+] = \sqrt{(10^{-4.80})(0.1470)} = 2.329 \times 10^{-6} \text{ mol dm}^{-3}$ <p><math>\text{pH} = 5.63 \quad [1\text{m}]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|  |  |      |                                                                                                                                                                                                                                                                                                                                                                |
|--|--|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | (iv) | Suggest and explain a suitable indicator for the above titration.                                                                                                                                                                                                                                                                                              |
|  |  |      | <u>Phenolphthalein</u> is a suitable indicator. [1m] This is a <u>weak acid-strong base titration</u> . The <u>pH range of the indicator (8-10) falls within the pH change at equivalence point (7-11)</u> [1m]                                                                                                                                                |
|  |  | (v)  | With the aid of an equation, explain why the pH of the titration curve at region X remains fairly constant. [6]                                                                                                                                                                                                                                                |
|  |  |      | When base is added, the added OH <sup>-</sup> is removed by reacting with the large reservoir of 2-methylbutanoic acid molecules present.<br><br>$\text{OH}^- + \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2\text{H} \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CO}_2^-\text{Na}^+ + \text{H}_2\text{O}$ [1m equation & explain] |
|  |  |      | [Total: 10]                                                                                                                                                                                                                                                                                                                                                    |

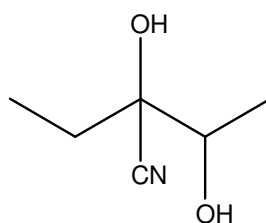
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                           |  |
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| 3 | <p>2-methylbutane is a component of natural gas. It is used as a fuel and also as an active ingredient in toothpaste for sensitive teeth.</p> <p>2-methylbutane (Mr = 72.0) can be synthesised from 3-methylbut-1-ene via a reaction with hydrogen under suitable conditions. The enthalpy change of this reaction is <math>-116 \text{ kJ mol}^{-1}</math>.</p> <div style="text-align: center;">  </div> |                                                                                                                                                                                                                                                                                                                                           |  |
|   | (i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Suggest the conditions for this reaction.                                                                                                                                                                                                                                                                                                 |  |
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Ni/Heat or Pt/rt or Pd/rt [1m]                                                                                                                                                                                                                                                                                                            |  |
|   | (ii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Calculate the enthalpy change of combustion of 2-methylbutane given the following enthalpy changes:                                                                                                                                                                                                                                       |  |
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) \quad \Delta H_1 = -125 \text{ kJ mol}^{-1}$ $(\text{CH}_3)_2\text{CHCH}=\text{CH}_2(\text{g}) + \frac{15}{2}\text{O}_2 \rightarrow 5\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{l}) \quad \Delta H_2 = -3503 \text{ kJ mol}^{-1}$ |  |
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\Delta H_{\text{rxn}} = \sum \Delta H_{\text{c}}(\text{reactants}) - \sum \Delta H_{\text{c}}(\text{products})$ $-116 = (-125) + (-3503) - \Delta H_{\text{c}}(\text{2-methylbutane}) \text{ [1m]}$ $\Delta H_{\text{c}}(\text{2-methylbutane}) = -3510 \text{ kJ mol}^{-1} \text{ (3sf) [1m]}$                                          |  |

|  |     |       |                                                                                                                                                                                                                                                                                                      |
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|  |     | (iii) | Calculate the minimum mass of 2-methylbutane required to heat 1000 g of water from 25 °C to boiling point. The specific heat capacity of water is 4.2 J g <sup>-1</sup> K <sup>-1</sup> . [5]                                                                                                        |
|  |     |       | Heat absorbed by water = $mc\Delta T = (1000)(4.2)(100-25) = 315000 \text{ J}$ [1m]<br><br>$\Delta H = -Q/n$<br>$(-3510 \times 1000) = -315000/(m/72)$<br>$m = 6.46 \text{ g (3sf)}$ [1m]                                                                                                            |
|  | (b) |       | Refer to the reaction scheme of the synthesis of 2-methylbut-2-ene from 2-methylbutane below.                                                                                                                                                                                                        |
|  |     |       |                                                                                                                                                                                                                                                                                                      |
|  |     | (i)   | Draw the structure of a possible intermediate for the reaction in the space above.                                                                                                                                                                                                                   |
|  |     |       | <br><div style="display: flex; align-items: center; justify-content: center;"> <div style="text-align: center;"> <br/>Cl         </div> <div style="margin: 0 10px;">or</div> <div style="text-align: center;"> <br/>Cl         </div> </div> <div style="text-align: right; color: red;">[1m]</div> |
|  |     | (ii)  | State the type of reaction occurring in steps 1 and 2.                                                                                                                                                                                                                                               |
|  |     |       | Step 1: substitution Step 2: elimination [1m for both]                                                                                                                                                                                                                                               |
|  |     | (iii) | Give reagents and conditions for steps 1 and 2.                                                                                                                                                                                                                                                      |
|  |     |       | Step 1: (limited) Cl <sub>2</sub> , UV light [1m]<br>Step 2: ethanolic NaOH, heat under reflux [1m]                                                                                                                                                                                                  |
|  |     | (iv)  | Will the yield of 2-methylbut-2-ene be high? Explain your answer. [5]                                                                                                                                                                                                                                |
|  |     |       | No. Substitution of alkanes is uncontrollable and there will be a mixture of various substitution products. [1m]                                                                                                                                                                                     |
|  |     |       | [Total: 10]                                                                                                                                                                                                                                                                                          |

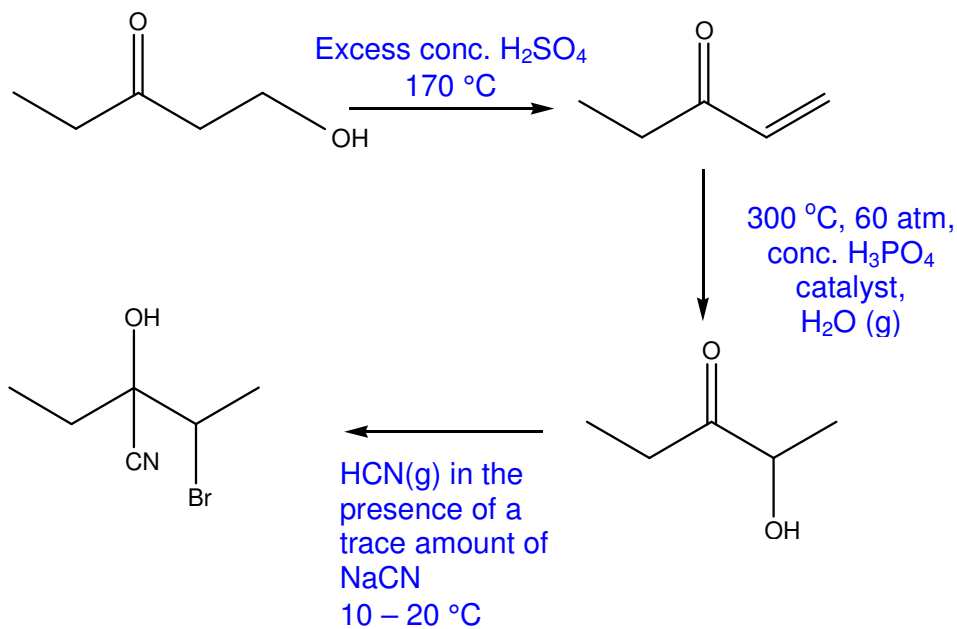
- 4** Pheromones are compounds an animal produces which can influence a change in the behavior of another animal or insect of the same species. Compound **A** is an isomer of an insect pheromone with a distinct peanut and hay-like odour characteristic.

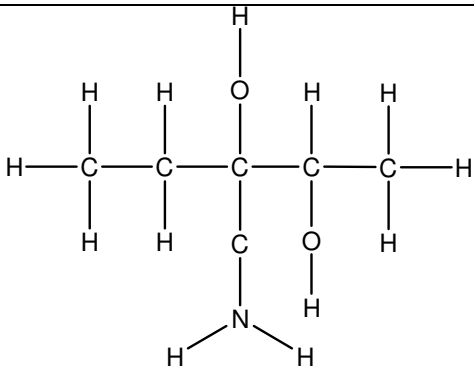
Compound **A**

- (a)** In not more than 3 steps, suggest a synthetic route for the formation of **B** from **A**. State clearly the reagents and conditions for each reaction step.

Compound **B**

[5]



|  |     |                                                                                                                                                                                                                                                                                                                           |
|--|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | (b) | Compound <b>B</b> can undergo a single reaction to form an amine. Draw the displayed formula of this amine and state the type of reaction and the reagents and conditions used in this reaction. [3]                                                                                                                      |
|  |     |  <p>Type of reaction: Reduction<br/>Reagents and conditions: LiAlH<sub>4</sub> in dry ether</p>                                                                                                                                          |
|  | (c) | Suggest the reagents and conditions for a chemical test that could be used to distinguish between compounds <b>A</b> and <b>B</b> . [2]                                                                                                                                                                                   |
|  |     | <p>Test: NaOH (aq), I<sub>2</sub>, warm<br/>Observations with A: A yellow ppt is observed.<br/>Observations with B: A yellow ppt is not observed.</p> <p>OR</p> <p>Test: 2,4-dinitrophenylhydrazine, warm<br/>Observations with A: An orange ppt is observed.<br/>Observations with B: An orange ppt is not observed.</p> |
|  |     | [Total: 10]                                                                                                                                                                                                                                                                                                               |



## Section B

Answer **two** questions from this section on separate answer paper.

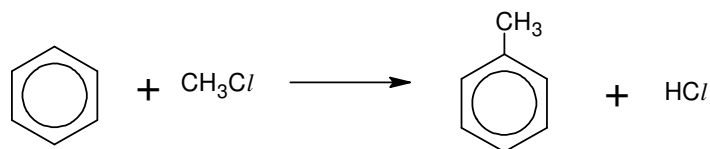
|   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | (a) | Sketch a graph to show how ionic radius changes across period 3 elements and explain the shape of your graph. [3]                                                                                                                                                                                                                                                                                                                                                                                                                             |
|   |     | <div data-bbox="354 457 792 814"> <p style="text-align: center;">[1]</p> </div> <p>Anions have <u>an additional shell of electrons</u> than cations of the same period thus have larger radii. [1]</p> <p>Across the period, ionic radius decreases for cations and anions (ions which are isoelectronic) as <u>nuclear charge increase while shielding effect remains the same</u>. Hence <u>effective nuclear charge increases</u> and the outermost electron shell is pulled slightly closer, giving rise to the decreasing trend. [1]</p> |
|   | (b) | (i) With the aid of an equation, define the 2 <sup>nd</sup> ionisation energy of gallium.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|   |     | $\text{Ga}^+(g) \rightarrow \text{Ga}^{2+}(g) + e^-$ [1m]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|   |     | The graph below shows the 1 <sup>st</sup> 8 ionisation energies of element D.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|   |     | <div data-bbox="370 1417 1218 1837"> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | (ii)  | Prove with reasoning that Element <b>D</b> belongs to the same group as gallium in the Periodic Table. Write the electronic configuration of the outer shell of element <b>D</b> .                                                                                                                                                                                                                                                                                                                                            |
|     |       | The 4 <sup>th</sup> electron is from an inner quantum shell [1m] and it experiences increased electrostatic attraction with the nucleus. A large amount of energy is required to remove the 4 <sup>th</sup> electron. [1m]<br>$ns^2 np^1$ [1m]                                                                                                                                                                                                                                                                                |
|     | (iii) | Gallium(III) oxide is used as a precursor in the manufacturing of semiconductor materials. Draw the dot-and-cross diagram of gallium(III) oxide. [6]                                                                                                                                                                                                                                                                                                                                                                          |
|     |       | $2 [Ga]^{3+} 3 [\ddot{O}]^{2-}$ <p>[2m; number of ions, charge on ions, number and position of valence electrons]</p>                                                                                                                                                                                                                                                                                                                                                                                                         |
| (c) |       | Period 3 oxides can be formed by combustion of the elements in oxygen. Describe the formation of sodium oxide and phosphorus pentoxide and write down the relevant equations of the reactions. [3]                                                                                                                                                                                                                                                                                                                            |
|     |       | $2 Na (s) + \frac{1}{2} O_2 (g) \rightarrow Na_2O (s)$<br><u>Sodium burns vigorously with an intense yellow flame to form a white ionic solid of sodium oxide.</u> [1m]<br>$P_4 (s) + 5 O_2 (g) \rightarrow P_4O_{10} (s)$ [1m for both]<br><u>Phosphorus reacts rapidly with <math>O_2</math> with a pale buish green flame to form a white <math>P_4O_{10}</math> solid.</u> [1m]                                                                                                                                           |
| (d) |       | <p>The addition of <math>Cl_2</math> to water forms hydrochloric acid and hypochlorous acid according to the following equation.</p> <p style="text-align: center;">Reaction I: <math>Cl_2 + H_2O \rightleftharpoons HCl + HClO</math></p> <p>Hypochlorous acid hydrolyses in water to form a hypochlorite ion, <math>OCl^-</math>. The hypochlorite ion formed is the main component in bleach and disinfectant agents.</p> <p style="text-align: center;">Reaction II: <math>HClO \rightleftharpoons OCl^- + H^+</math></p> |
|     | (i)   | Explain why the formation of the stable hypochlorite ion is facilitated by dissolving chlorine gas in basic solutions, such as sodium hydroxide.                                                                                                                                                                                                                                                                                                                                                                              |
|     |       | <p><u>NaOH will react with HCl, reducing the concentration of HCl. The position of equilibrium of reaction 1 will shift to the right to increase the [HCl] and increases the [HClO].</u> [1m]</p> <p><u>When [HClO] increases, the position of equilibrium will shift to the right in reaction 2 to reduce the [HClO] and forms <math>OCl^-</math>. This stabilises the hypochlorite in NaOH.</u> [1m]</p>                                                                                                                    |

|                          |                             | (ii)  | 1 mole of $\text{Cl}_2$ was dissolved in $1 \text{ dm}^3$ of water with a dissociation coefficient of 0.75 in reaction I. Calculate the value of $K_c$ , giving its units.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                             |                          |     |                          |       |       |       |             |      |      |      |
|--------------------------|-----------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------|--------------------------|-----|--------------------------|-------|-------|-------|-------------|------|------|------|
|                          |                             |       | $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HClO}$ <table><tr><td>Initial</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Change</td><td>-0.75</td><td>+0.75</td><td>+0.75</td></tr><tr><td>Equilibrium</td><td>0.25</td><td>0.75</td><td>0.75</td></tr></table> <p>[1m for equilibrium values]</p> $K_c = \frac{\left(\frac{0.75}{1}\right)\left(\frac{0.75}{1}\right)}{\left(\frac{0.25}{1}\right)} = 2.25 \text{ mol dm}^{-3}$ <p>[1m for correct <math>K_c</math> expression, 1m for values and units]</p>                                                                                               | Initial  | 1                           | 0                        | 0   | Change                   | -0.75 | +0.75 | +0.75 | Equilibrium | 0.25 | 0.75 | 0.75 |
| Initial                  | 1                           | 0     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |                             |                          |     |                          |       |       |       |             |      |      |      |
| Change                   | -0.75                       | +0.75 | +0.75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |                             |                          |     |                          |       |       |       |             |      |      |      |
| Equilibrium              | 0.25                        | 0.75  | 0.75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                             |                          |     |                          |       |       |       |             |      |      |      |
|                          |                             | (iii) | <p>There are two unlabelled solutions <b>X</b> and <b>Y</b>, each containing the same concentration of <math>\text{HCl}</math> (aq) and <math>\text{HOCl}</math> (aq). To identify the solutions a student mixed equal volumes of <math>\text{NaOH}</math> (aq) with each solution. The following results are obtained:</p> <table><tr><th>Reaction</th><th><math>\Delta T / ^\circ\text{C}</math></th></tr><tr><td><math>\text{NaOH} + \text{X}</math></td><td>4.8</td></tr><tr><td><math>\text{NaOH} + \text{Y}</math></td><td>5.6</td></tr></table> <p>Deduce the acid present in solutions X and Y and explain your answer. [8]</p> | Reaction | $\Delta T / ^\circ\text{C}$ | $\text{NaOH} + \text{X}$ | 4.8 | $\text{NaOH} + \text{Y}$ | 5.6   |       |       |             |      |      |      |
| Reaction                 | $\Delta T / ^\circ\text{C}$ |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |                             |                          |     |                          |       |       |       |             |      |      |      |
| $\text{NaOH} + \text{X}$ | 4.8                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |                             |                          |     |                          |       |       |       |             |      |      |      |
| $\text{NaOH} + \text{Y}$ | 5.6                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |                             |                          |     |                          |       |       |       |             |      |      |      |
|                          |                             |       | <p>X contains <math>\text{HOCl}</math>. Y contains <math>\text{HCl}</math>. [1m]</p> <p>In the neutralisation reactions, the enthalpy change of reaction of the weak acid is <u>less exothermic</u> [1m] than that of a strong acid as it will <u>involve the enthalpy change of dissociation of <math>\text{H}^+</math></u> [1m] from the weak acid as it undergoes partial dissociation. Energy is required for the dissociation.</p>                                                                                                                                                                                                 |          |                             |                          |     |                          |       |       |       |             |      |      |      |
|                          |                             |       | [Total: 20]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                             |                          |     |                          |       |       |       |             |      |      |      |

|   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | (a) | <p>Methylbenzene can become 2-chlorobenzoic acid and 3-chlorobenzoic acid via different reaction synthesis. Give the reagents and conditions required for the synthesis of 2-chlorobenzoic acid, write the chemical equations and name the type of reactions that occur. State and explain how 3-chlorobenzoic acid may be synthesised instead of 2-chlorobenzoic acid. [6]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|   |     | <p>Step 1: Substitution of methylbenzene to form 2-chloromethylbenzene.</p> <p>Reagent &amp; Condition: <math>\text{Cl}_2</math> (l) with anhydrous <math>\text{FeCl}_3</math></p> <div data-bbox="506 583 1203 737"> <math display="block">\text{C}_6\text{H}_5\text{CH}_3 + \text{Cl}_2 \longrightarrow \text{C}_6\text{H}_4(\text{CH}_3)\text{Cl} + \text{HCl}</math> </div> <p>[2m for name of reaction, reagent &amp; condition and equation]</p> <p>Step 2: Oxidation of 2-chloromethylbenzene to form 2-chlorobenzoic acid.</p> <p>Reagent &amp; Condition: <math>\text{KMnO}_4</math> (aq), <math>\text{H}_2\text{SO}_4</math> (aq), Heat under reflux.</p> <div data-bbox="597 1087 1179 1241"> <math display="block">\text{C}_6\text{H}_4(\text{CH}_3)\text{Cl} \xrightarrow{3[\text{O}]} \text{C}_6\text{H}_4(\text{COOH})\text{Cl} + \text{H}_2\text{O}</math> </div> <p>[2m for name of reaction, reagent &amp; condition and equation]</p> <p>3-chlorobenzoic acid can be synthesised from methylbenzene by changing the sequence of reactions; <u>oxidation followed by substitution</u>. [1m]</p> <p>The products are different as <u>different existing substituent affects the position of the incoming substituent</u>. [1m] Methyl substituent is 2,4-directing while carboxylic acid substituent is 3,5-directing.</p> |
|   | (b) | <p>Explain why 2-chlorobenzoic acid is a stronger acid than benzoic acid. [2]</p> <p>The electronegative Cl atom of chlorobenzoic acid will exert an electron-withdrawing effect on the corresponding carboxylate anion. [1m]</p> <p>The negative charge becomes more dispersed and the conjugate base becomes more stable. [1m]</p> <p>Hence the dissociation of chlorobenzoic acid to give <math>\text{H}^+</math> is more favoured.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

- (c) The equation for the formation of methylbenzene from benzene and chloromethane is shown.



A student performed an experiment to determine the order of reaction with respect to chloromethane. The initial concentration of chloromethane was  $0.0100 \text{ mol dm}^{-3}$  and benzene was used in large excess. The concentration of chloromethane at various times in the reaction mixture was found as follows:

| Time / s | [chloromethane] / $\text{mol dm}^{-3}$ |
|----------|----------------------------------------|
| 0        | 0.0100                                 |
| 40       | 0.0079                                 |
| 80       | 0.0062                                 |
| 120      | 0.0049                                 |
| 160      | 0.0038                                 |
| 200      | 0.0030                                 |

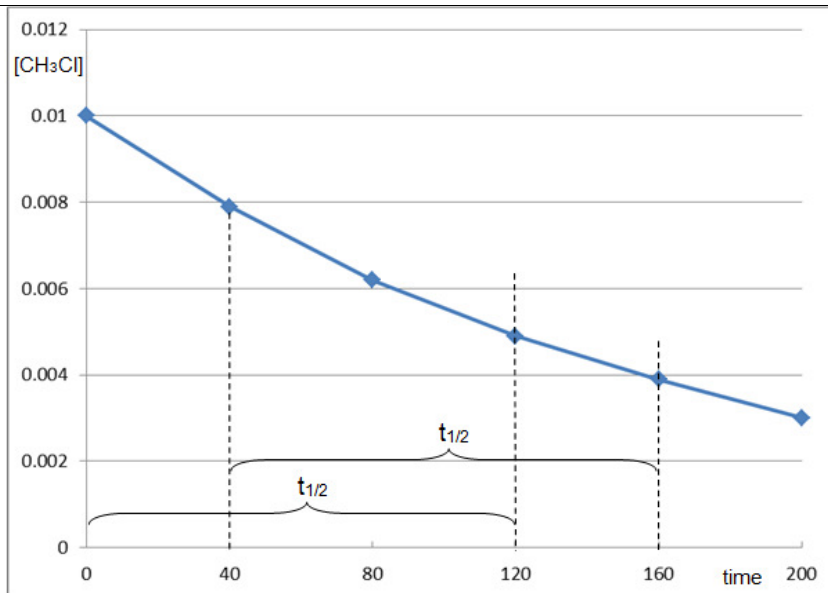
- (i) Explain what is meant by *order of reaction*.

The term “order of reaction” refers to the power to which the concentration of the reactant is raised in the experimentally determined rate equation. [1m]

Or

For a general rate law,  $\text{rate} = k[\text{A}]^m$  experimentally determined  
 [A] is the concentration of the reactant A,  
 m is the order of the reaction with respect to the reactant A

- (ii) Plot the above data on suitable axes and use your graph to determine the order of reaction with respect to chloromethane.



[1m for points plotted accurately and smooth curved line drawn, 1m for axes labeled correctly, 1m for showing clearly on the graph dotted lines to find two  $t_{1/2}$ ]

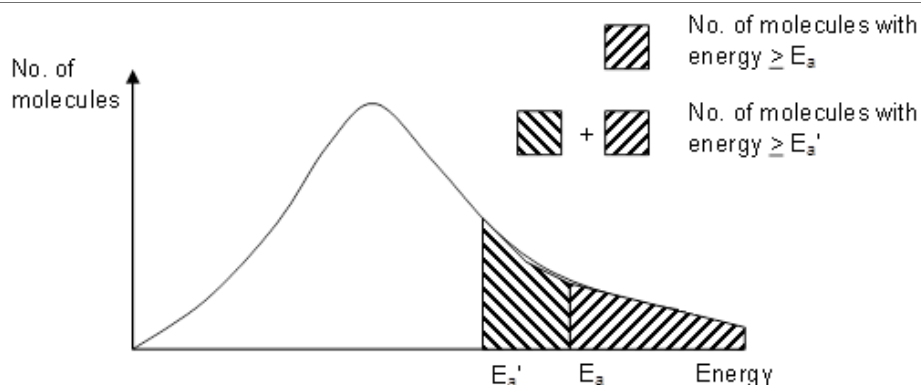
From the graph, half-lives are approximately constant at 120s. [1m]

Hence, order of reaction with respect to  $\text{CH}_3\text{Cl}$  is 1. [1m]

(iii) The initial concentration of chloromethane was  $0.0100 \text{ mol dm}^{-3}$  and benzene was used in large excess. Suggest why benzene was used in large excess in this experiment?

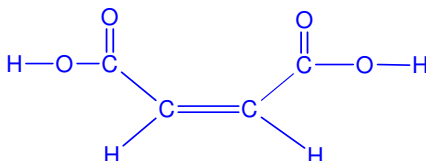
Benzene was used in large excess so that its concentration will remain almost constant throughout the experiment. As such, the rate of reaction will only depend on chloromethane concentration [1m] thus enabling the determination of the order of reaction with respect to chloromethane. [1m]

(iv) Explain with the aid of a suitable diagram, how the presence of a catalyst can cause an increase in the rate of a chemical reaction. [12]



[1m for correct shape and axes, 1m for labelling and legend]

|  |  |  |                                                                                                                                                                                                                                                                                                          |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | In the presence of a catalyst, reaction has a <u>different mechanism with a lower activation energy</u> compared to the uncatalysed reaction. [1m]<br><u>More molecules will possess energy greater than this lowered activation energy, hence frequency of effective collisions will increase.</u> [1m] |
|  |  |  | [Total: 20]                                                                                                                                                                                                                                                                                              |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                               |  |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 7   | Maleic acid is one of two stereoisomers of butenedioic acid, HOOC(CH) <sub>2</sub> COOH. Maleic acid is the <i>cis</i> -isomer whereas fumaric acid is the <i>trans</i> -isomer. This question examines the chemistry of maleic acid and its derivatives.                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                               |  |
| (a) | (i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Draw the full structural formula of maleic acid.                                                                                                                                                                                                                                                                                              |  |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <div></div> <p>Must be cis and trigonal planar wrt C [1m]<br/>Full structural/displayed formula [1m]</p>                                                                                                                                                    |  |
|     | (ii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The two stereoisomers have different melting points as intramolecular hydrogen bonding is possible only in one isomer. Identify this isomer and hence, explain whether it has a higher or lower melting point, compared to the other isomer. [4]                                                                                              |  |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Maleic acid [1m] has intramolecular hydrogen bonding (as the two COOH groups are closer to each other).</p> <p><u>Maleic acid should have a lower melting point</u> as it can form <u>less extensive intermolecular hydrogen bonds</u> (between molecules) hence <u>less energy is required to break the weak hydrogen bonds.</u> [1m]</p> |  |
| (b) | Maleic acid is not an approved additive as long term consumption of high levels of maleic acid can cause kidney damage. In May 2013, the Agri-food & Veterinary Authority of Singapore recalled several food products imported from Taiwan that were found to contain maleic acid. Most of the affected products were tapioca starch balls used in the preparation of bubble tea beverages.                                                                            |                                                                                                                                                                                                                                                                                                                                               |  |
|     | Some physical and chemical properties of maleic acid are given below.                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                               |  |
|     | <div><p><b>Molecular formula:</b> C<sub>4</sub>H<sub>4</sub>O<sub>4</sub><br/><b>Physical state and appearance:</b> Powdered solid<br/><b>Colour:</b> White.<br/><b>M<sub>r</sub>:</b> 116.08<br/><b>Density:</b> 1.59 g cm<sup>-3</sup><br/><b>Acid/Base properties:</b> Dibasic<br/><b>Water/Oil Distribution Coefficient:</b> The product is more soluble in water;<br/><math display="block">\lg \frac{[\text{in oil}]}{[\text{in water}]} = -0.5</math></p></div> |                                                                                                                                                                                                                                                                                                                                               |  |

|  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |      | In a volumetric analysis to determine percentage by mass of maleic acid present in tapioca starch balls, a 5.0 g sample was first powdered and dissolved in 100.0 cm <sup>3</sup> of an oil-based solvent. Most of the maleic acid was extracted from the oil-based solvent into an aqueous medium with 100.0 cm <sup>3</sup> distilled water. 25.0 cm <sup>3</sup> of the aqueous solution required 7.10 cm <sup>3</sup> of 0.05 mol dm <sup>-3</sup> aqueous sodium hydroxide for complete neutralisation.                                                                                                   |
|  | (i)  | Using the data provided, determine the concentration of maleic acid present in 100.0 cm <sup>3</sup> of aqueous solution. You may assume the only acid present in tapioca starch balls is maleic acid.                                                                                                                                                                                                                                                                                                                                                                                                         |
|  |      | $n_{\text{NaOH}} = \frac{7.10}{1000} \times 0.05$ $= 3.550 \times 10^{-4} \text{ mol}$ <p>Since maleic acid is diprotic,</p> $n_{\text{acid}} \text{ in } 25.0 \text{ cm}^3 = \frac{1}{2} \times 3.550 \times 10^{-4}$ $= 1.775 \times 10^{-4} \text{ mol [1m]}$ $n_{\text{acid}} \text{ in } 100.0 \text{ cm}^3 = \frac{100.0}{25.0} \times 1.775 \times 10^{-4}$ $= 7.100 \times 10^{-4} \text{ mol [ecf 1m]}$ $[\text{acid}] = \frac{7.100 \times 10^{-4}}{0.100} = 7.100 \times 10^{-3} \text{ mol dm}^{-3} \text{ [ecf 1m]}$ <p>*If student just find concentration in 25.0cm<sup>3</sup> sample [2m]</p> |
|  | (ii) | <p>When a solute is added to two solvents, A and B, which do not mix, some of the solute dissolves in each of the solvents and an equilibrium is set up between the two solvents. It has been shown that for dilute solutions, the ratio of the two concentrations in each of the solvents at equilibrium is a constant known as the <b>Dissociation Coefficient</b>.</p> <p>Using your answer in (i) and the data provided, determine the concentration of maleic acid remaining in the oil-based solvent after extraction.</p>                                                                               |
|  |      | $\lg \frac{[\text{in oil}]}{[\text{in water}]} = -0.5$ $\lg \frac{[\text{in oil}]}{[7.100 \times 10^{-3}]} = -0.5$ $\frac{[\text{in oil}]}{[7.100 \times 10^{-2}]} = 10^{-0.5}$ $[\text{in oil}] = 2.245 \times 10^{-3} \text{ mol dm}^{-3} \text{ [ecf1m]}$                                                                                                                                                                                                                                                                                                                                                   |



|     |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |      | (iii) | Hence, using your answer in (i) and (ii) and data provided, determine the percentage by mass of maleic acid in the sample of tapioca starch balls.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|     |      |       | [7]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|     |      |       | $n_{\text{acid total}} = n_{\text{acid (in water)}} + n_{\text{acid (in oil)}}$ $= 7.100 \times 10^{-3} \times \frac{100}{1000} + 2.245 \times 10^{-3} \times \frac{100}{1000} \text{ [ecf1m]}$ $= 9.345 \times 10^{-4} \text{ mol}$ $m_{\text{acid}} = 9.345 \times 10^{-4} \times 116.08 \text{ [ecf1m]}$ $= 0.108 \text{ g}$ $\% \text{ mass} = \frac{0.108}{5.0} \times 100\% = 2.17\% \text{ [ecf1m]}$                                                                                                                                                                                    |
| (c) | (i)  |       | Maleic acid can be reduced to a compound <b>S</b> with molecular formula $\text{C}_4\text{H}_8\text{O}_2$ . State the reagents required for this conversion and draw the full structural formula of <b>S</b> .                                                                                                                                                                                                                                                                                                                                                                                 |
|     |      |       | <p>Reagents and conditions: <math>\text{LiAlH}_4</math> in dry ether [1m]</p> <p><b>S:</b></p> $  \begin{array}{ccccccc}  & \text{H}-\text{O} & & & & \text{O}-\text{H} & \\  &   & & & &   & \\  \text{H} & -\text{C} & - & \text{C} & = & \text{C} & -\text{C}-\text{H} \\  &   & &   & &   & \\  & \text{H} & & \text{H} & & \text{H} &   \end{array}  $ <p>[1m]</p>                                                                                                                                                                                                                        |
|     | (ii) |       | <p>Compound <b>S</b> has other structural isomers <b>T</b> and <b>U</b> all having the molecular formula <math>\text{C}_4\text{H}_8\text{O}_2</math>.</p> <p>1 mole of <b>T</b> gives 1 mole of yellow precipitate with aqueous alkaline iodine. <b>T</b> also decolourises aqueous bromine and exhibits geometric isomerism.</p> <p>1 mole of <b>U</b> also gives 1 moles of yellow precipitate with aqueous alkaline iodine. However <b>U</b> does not decolourise aqueous bromine.</p> <p>Deduce the structures of compounds <b>T</b> and <b>U</b>. Explain your reasoning clearly. [9]</p> |
|     |      |       | <p><b>T:</b></p> $  \begin{array}{ccccccc}  & \text{OH} & \text{H} & & \text{OH} & & \\  &   &   & &   & & \\  \text{H} & -\text{C} & = & \text{C} & - & \text{C} & -\text{CH}_3 \\  & & & & &   & \\  & & & & & \text{H} &   \end{array}  $ <p><b>U:</b></p> $  \begin{array}{ccccccc}  & \text{OH} & \text{H} & & \text{O} & & \\  &   &   & &    & & \\  \text{H} & -\text{C} & - & \text{C} & - & \text{C} & -\text{CH}_3 \\  &   &   & & & & \\  & \text{H} & \text{H} & & & &   \end{array}  $ <p>[2m]</p>                                                                               |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p><b>Deductions</b></p> <p> <math display="block">\begin{array}{c} \text{OH} \\   \\ \text{C}-\text{CH}_3 \\   \\ \text{H} \end{array}</math>           in <b>T</b> undergoes <u>oxidation</u> with <u>aq I<sub>2</sub></u> to form a yellow ppt. [1m]         </p> <p>alkene in <b>T</b> undergoes <u>addition</u> with <u>aq Br<sub>2</sub></u>. [1m]</p> <p><b>T</b> exhibits geometric isomerism because each C in C=C is bonded to 2 different groups. [1m]</p> <p>alkene is absent in <b>U</b> since it does not react with aq Br<sub>2</sub>. [1m]</p> <p> <math display="block">\begin{array}{c} \text{O} \\    \\ \text{C}-\text{CH}_3 \end{array}</math>           in <b>U</b> undergoes <u>oxidation</u> with <u>aq I<sub>2</sub></u> to form a yellow ppt. [1m]         </p> <p>           Since only 1 mole of CHI<sub>3</sub> is formed,           <math display="block">\begin{array}{c} \text{OH} \\   \\ \text{C}-\text{CH}_3 \\   \\ \text{H} \end{array}</math>           cannot also be present in <b>U</b> [1m], hence alcohol should be on the terminal carbon.         </p> <p>*6 Marking points but MAX 5 for deductions.</p> <p>*If students write methyl alcohol or methyl carbonyl present in <b>T/U</b>... give [1m]</p> |
|  |  |  | [Total: 20]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |